

Introduction

Induction is a critical technique in mathematics (and computer science) which allows us to quickly and easily prove statements that are true on all of $\mathbb{Z}^+$.

Learning Objectives

- Explain how mathematical induction works and what it is designed to prove.
- Prove basic statements by using mathematical induction.
- Expand a sum or product from sigma/pi-notation to an explicit finite sum/product and vice versa.
- Prove basic summation formulas using induction.
- Explain the difference between strong and regular induction. Use strong induction to do elementary proofs.
- Explain what it means for something to be defined recursively.
- Use (structural) induction to prove properties of recursively defined elements.

The Principle of Mathematical Induction

Suppose P is a predicate and you wish to prove $P(n)$ is true for all $n \in \mathbb{Z}^+$. How do you do this? Mathematical induction operates on the domino principle, by creating a chain of implications which extends to every natural number. For example, suppose that we want to show that $P(n)$ is true for every $n \in \mathbb{Z}^+$. If we can show that $P(1)$ is true and that the implication $P(k) \Rightarrow P(k+1)$ is true for every $k \in \mathbb{Z}^+$, then the result holds for any n . This is precisely because

$$P(1) \Rightarrow P(2) \Rightarrow P(3) \Rightarrow \cdots \Rightarrow P(n-1) \Rightarrow P(n) \Rightarrow \cdots,$$

and since $P(1)$ is true, so too is every $P(n)$ thereafter.

Mathematical Induction

Let P be some predicate. If $P(1)$ is true and $P(k) \Rightarrow P(k+1)$ for any k , then $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Exercise 1 Induction and the Well-Ordering Principle are logically equivalent; that is, if you start with the Well-Ordering Principle, you can prove Induction. Conversely, if you start with Induction, you can prove the Well-Ordering Principle. The proofs are not too difficult. See if you can prove the equivalence!

Thus mathematical induction consists of two steps. The first is to demonstrate the *base case* that $P(1)$ is true. The second is to invoke the *induction hypothesis* that $P(k)$ is true for some k , and demonstrate that $P(k) \Rightarrow P(k+1)$.

Example 2

Show, using mathematical induction, that $2n + 2 \leq 4n$ for all integers $n \geq 1$.

Solution.

1. **Base Case:** The smallest number for which this occurs is $n = 1$, and in this case we have $2n + 2 = 4$ and $4n = 4$, so the result holds in the base case.
2. **Induction Step:** Assume that $2k + 2 \leq 4k$ for some natural number k . We want to show that $2(k + 1) + 2 \leq 4(k + 1)$. Indeed, notice that

$$\begin{aligned}
 4(k + 1) &= 4k + 4 \\
 &\geq (2k + 2) + 4 && \text{using the induction hypothesis } 4k \geq 2k + 2 \\
 &\geq 2k + 2 + 2 && \text{since } 4 \geq 2 \\
 &= 2(k + 1) + 2.
 \end{aligned}$$

We conclude from the induction principle that $2k + 2 \leq 4k$ for all $k \in \mathbb{Z}^+$. ■

Exercise 3 Show that for every $n \in \mathbb{Z}^+$, $2n \leq 2^n$.

Example 4

Prove that for all positive integers k , $5 \mid (6^k - 1)$.

Solution.

1. **Base Case:** The simplest case is $k = 1$, from which we get that $6^k - 1 = 5$ and clearly $5 \mid 5$.
2. **Induction Step:** For some positive integer k , assume that $5 \mid (6^k - 1)$. Let $d \in \mathbb{Z}$ be such that $5d = 6^k - 1$. To show the result for the $(k + 1)$ -case, write $6^{k+1} - 1$ as

$$\begin{aligned}
 6^{k+1} - 1 &= 6(6^k) - 1 \\
 &= (1 + 5)(6^k) - 1 \\
 &= 5(6^k) + (6^k - 1) \\
 &= 5(6^k + d) && \text{since } 5d = 6^k - 1.
 \end{aligned}$$

Hence $5 \mid (6^{k+1} - 1)$ as required. ■

Exercise 5: Bernoulli's Inequality Show that for all $x \geq -1$ and $n \in \mathbb{Z}^+$, we have

$$(1 + x)^n \geq 1 + nx.$$

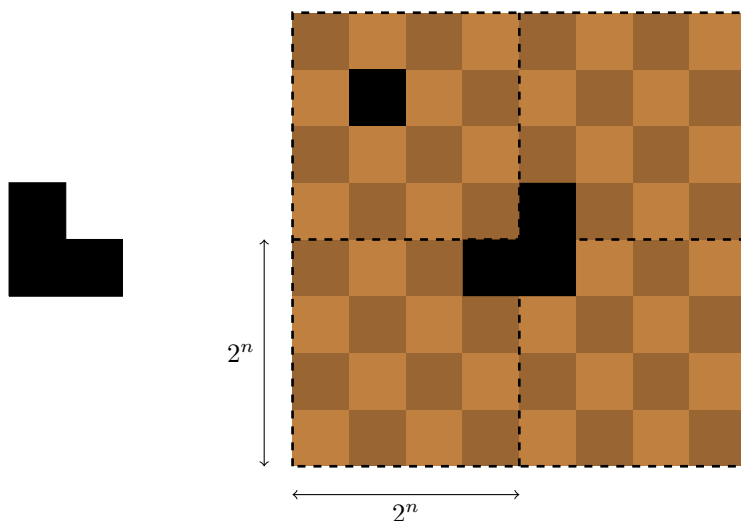


Figure 1: Left: The base case. A 2×2 board with a single tile removed is an L -shape, and so admits an L -tiling. Right: The induction step. By placing the first tile as such, each of the quadrants is a $2^n \times 2^n$ board with a single tile removed.

Example 6

Show that for all $n \in \mathbb{Z}^+$, a $2^n \times 2^n$ chessboard with a single tile removed can be L -tiled; that is, tiled by an L -shape consisting of three squares.

Solution. The base case is when $n = 1$, in which we are tasked with tiling a 2×2 chessboard with one tile removed. This is immediately possible, so we are done.

Assume then that any $2^n \times 2^n$ board with a single tile removed admits an L -tiling, and consider a $2^{n+1} \times 2^{n+1}$ board. Divide this board into four quarters, so that each quarter has dimension $2^n \times 2^n$. By rotating the board if necessary, assume that the missing tile is located in the upper-left quadrant. We place our first tile as illustrated in Figure 1. Note that, excluding the placement of the first tile, every quadrant is now a $2^n \times 2^n$ board with a single tile removed. By the induction hypothesis, each board admits an L -tiling, so those tilings combined give the tiling of the $2^{n+1} \times 2^{n+1}$ board. ■

Exercise 7 Suppose you are given a $2 \times n$ grid of squares and a collection of dominoes that cover a 2×1 area. Dominoes can be laid horizontally or vertically, but not diagonally; nor may dominoes overlap. Determine how many tilings of the $2 \times n$ grid are possible, and prove your result. See Figure 2 for the set of all 2×3 tilings.

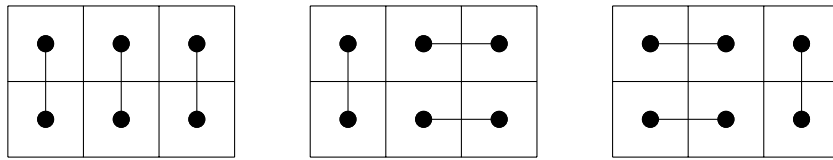


Figure 2: The possible domino tilings of the 2×3 grid.

Summation (Sigma) Notation

Sigma notation is used to make complicated sums much easier to write down. In particular, we use a summation index to iterate through elements of a list and then sum them together. Consider the expression

$$\sum_{i=n}^m r_i \quad (1)$$

which is read as “the sum from $i = n$ to m of r_i .” The element i is known as the *dummy* or *summation* index, n and m are known as the *summation bounds*, and r_i is the *summand*. In order to decipher this rather cryptic notation, we adhere to the following algorithm:

1. Set $i = n$ and write down r_i ;
2. Add 1 to the index i and add r_i to the current sum;
3. If i is equal to m then stop, otherwise go to step 2 and repeat.

For those computer savvy students out there, this is nothing more than a for-loop. Interpreting (1) we thus have

$$\sum_{i=n}^m r_i = r_n + r_{n+1} + r_{n+2} + \cdots + r_m.$$

Example 8

Let k be some fixed positive integer such that $k \geq 1$. Show that

$$\sum_{n=1}^k \frac{1}{n(n+1)} = \frac{k}{k+1}.$$

Solution. We proceed by induction

1. **Base Case:** When $k = 1$ the left-hand-side becomes

$$\sum_{n=1}^1 \frac{1}{n(n+1)} = \frac{1}{1(1+1)} = \frac{1}{2}.$$

This agrees with the right-hand-side $1/(1+1) = 1/2$, so the base case is satisfied.

2. Induction Step: Let k be some fixed arbitrary positive integer and assume that

$$\sum_{n=1}^k \frac{1}{n(n+1)} = \frac{k}{k+1}.$$

We would like to show that the result holds for $k+1$. It makes most sense to start by working with the left-hand-side, since it will give us the most “flexibility.” Notice that

$$\begin{aligned} \sum_{n=1}^{k+1} \frac{1}{n(n+1)} &= \frac{1}{n(n+1)} \Big|_{n=k+1} + \sum_{n=1}^k \frac{1}{n(n+1)} \\ &= \frac{1}{(k+1)(k+2)} + \frac{k}{k+1} && \text{via the induction hypothesis} \\ &= \frac{1+k(k+2)}{(k+1)(k+2)} = \frac{k^2+2k+1}{(k+1)(k+2)} && \text{common denominator} \\ &= \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2} && \text{factoring and cancelling} \\ &= \frac{(k+1)}{(k+1)+1}. \end{aligned}$$

This is precisely what we wanted to show, and so we are done. ■

Exercise 9 Show that for any $k \geq 1$ we have

$$\sum_{n=1}^k n^2 = \frac{k(k+1)(2k+1)}{6}. \quad (2)$$

More General Induction

The principle of induction can be extended beyond just $\mathbb{Z}^+$. For example, suppose we want to show that the predicate $P(n)$ is true for all even numbers greater than or equal to 10. Here the base case is to demonstrate $P(10)$, followed by $P(2n) \Rightarrow P(2n+2)$. This creates the chain of implications

$$P(10) \Rightarrow P(12) \Rightarrow P(14) \Rightarrow \cdots \Rightarrow P(2n) \Rightarrow$$

demonstrating $P(n)$ for all even numbers greater than or equal to 10. This idea is easily generalized to any other induction scheme. Of course, this can be seen as equivalent to induction by setting $Q(n)$ to be the statement $P(2n)$.

An ostensibly different type of induction is that of *strong induction*. We again aim to create a chain of implications, but now we make a stronger induction hypothesis.

Theorem 10: Strong Induction

Let P be some predicate. Suppose that $P(1)$ is true, and moreover

$$P(1) \wedge P(2) \wedge \cdots \wedge P(k) \Rightarrow P(k+1)$$

for any k , then $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Proof. We will proceed by using (normal) induction. Let $Q(k) = P(1) \wedge \cdots \wedge P(k)$. Since $Q(1) = P(1)$, the base case is true. Now assume that $Q(n)$ is true. Since $Q(n) \Rightarrow P(n+1)$, then $P(n+1)$ is true and so $Q(n+1) = Q(n) \wedge P(n+1)$ is true as well. By induction, $Q(n)$ holds for all n , and this is only possible if all $P(n)$ are true, as required. $\square$

Hence (Induction) $\Rightarrow$ (Strong Induction). Moreover, since normal induction uses a weaker hypothesis, we see that (Strong Induction) $\Rightarrow$ (Induction). This shows that induction and strong induction are actually equivalent.

Example 11

Show that any postage amount greater than or equal to 8 cents can be formed by using 3 cent and 5 cent stamps.

Solution. Let $P(n)$ be the statement “A postage of n cents can be made of 3 and 5 cent stamps.” We will need to check three base cases for this proof (why?), which give

$$P(8) = 3(1) + 5(1), \quad P(9) = 3(3) + 5(0), \quad P(10) = 3(0) + 5(2).$$

Now assume that $P(k)$ is true for all $8 \leq k \leq n$, for which we will show that $P(n+1)$ is true. If $8 \leq k \leq 10$ then we’re done, so suppose that $n > 10$. In this case we can write $n+1 = (n-2) + 3$. By our induction hypothesis, we know that a postage of $n-2$ stamps can be resolved, say by r three-cent stamps and s five cent stamps, thus

$$n+1 = (n-2) + 3 = [3(r) + 5(s)] + 3 = 3(r+1) + 5(s)$$

as required. $\blacksquare$

Be sure to study Example 11 until you understand why three base cases were needed.

Exercise 12 Consider a two-player game, consisting of two bowls of marbles. Each player takes a turn removing any *positive* number of marbles from a *single* bowl. The player that removes the last marble wins. Show that if both bowls have an identical number of marbles, the player who goes second always has a winning strategy.

Exercise 13 Show that every positive integer has a binary representation; that is, every positive natural number can uniquely be written as a sum of distinct powers of 2.

Recursion and Structural Induction

Recursion is the process of creating a set by specifying how to construct new elements from existing elements in the set. Thus a recursively defined set always consists of

- *The Basis*: The set of elements which are assumed to be in the set initially.
- *The Constructor*: A (collection of) function(s) which indicates how new elements of the set are formed.

Implicit in this construction is that the only way for an element to belong to the set is to either be a basis element, or to have been constructed through the constructor.

For example, we can recursively construct a set E as follows:

- $0 \in E$
- If $e \in E$ then $-e \in E$ and $e + 2 \in E$.

If we wanted to formalize this, we could say that $b_0 = 0$ is our basis element and our constructors are $c_1(x) = -x$ and $c_2(x) = x + 2$. Thus if we start applying our constructor to the basis element, we would find that

$$c_1(0) = -0 \quad \text{and} \quad c_2(0) = 2$$

showing that $2 \in E$. Knowing that $\{0, 2\} \subseteq E$, we can continue constructing elements by applying the constructor to our newly found element 2:

$$c_1(2) = -2 \quad \text{and} \quad c_2(2) = 4.$$

Thus $\{-2, 0, 2, 4\} \subseteq E$. As we continue this process forever, we would eventually find that E is precisely the set of even integers, though this fact requires a proof.

We can define more than just sets using recursion as well. For example, the *Fibonacci sequence* is defined as follows: Set $x_1 = x_2 = 1$ and define $x_n = x_{n-1} + x_{n-2}$. Thus we have

$$x_3 = x_2 + x_1 = 2, \quad x_4 = x_3 + x_2 = 2 + 1 = 3, \quad x_5 = x_4 + x_3 = 3 + 2 = 5, \quad \dots$$

A sequence is not a set, since sequences keep track of the order of their elements. However, there are just too many different types of objects that can be defined recursively to give a single definition for recursion. Thus you should keep in mind that the idea is that new objects are defined by combining existing objects.

We can prove things about recursive elements by using induction!

Example 14

Consider the recursive sequence defined by $x_1 = 3, x_2 = 7$ and

$$x_k = 5x_{k-1} - 6x_{k-2}.$$

Show that $x_k = 2^k + 3^{k-1}$.

Solution. We will proceed by strong induction. As our construction of x_k refers to two previous elements, x_{k-1} and x_{k-2} , there are two base cases:

$$x_1 = 2^1 + 3^0 = 3, \quad x_2 = 2^2 + 3 = 7$$

which agrees with our result. Now assume that $x_k = 2^k + 3^{k-1}$ for all $1 \leq k \leq n$. Examining x_{n+1} we have

$$\begin{aligned} x_{n+1} &= 5x_n - 6x_{n-1} = 5(2^n + 3^{n-1}) - 6(2^{n-1} + 3^{n-2}) \\ &= 5(2^n + 3^{n-1}) - [3 \cdot 2^n + 2 \cdot 3^{n-1}] \\ &= 2 \cdot 2^n + 3 \cdot 3^{n-1} \\ &= 2^{n+1} + 3^n. \end{aligned}$$

exactly as desired. ■

Exercise 15 Let $x_1 = x_2 = 1$ and $x_n = x_{n-1} + x_{n-2}$ so that x_n is the Fibonacci sequence. Define

$$\alpha_{\pm} = \frac{1 \pm \sqrt{5}}{2}.$$

Show that $x_n = (\alpha_+^n - \alpha_-^n)/\sqrt{5}$.

The benefit of sequences is that there is an obvious “next” element to consider. However, recursively defined objects need not have that natural linear ordering, requiring a different type of induction.

Principle of Structural Induction

Suppose that S is a recursively defined set, with basis $b_1, \dots, b_n$ and constructor $c(b_1, \dots, b_k)$. If P is a predicate such that $P(b_1), \dots, P(b_n)$ is true, and for any $s_1, \dots, s_k \in S$ we have that $P(s_1) \wedge P(s_2) \wedge \dots \wedge P(s_k) \implies P(c(s_1, \dots, s_k))$ is true, then $P(s)$ holds for all $s \in S$.

Hence to prove that $P(s)$ is true for all $s \in S$, we need to show that the predicate is true for all of the basis cases, and show that it is true for each constructed element as well.

Example 16

Suppose that S is defined as follows:

- $3 \in S$
- If $a, b \in S$, then $a + b$ is in S .

Show that $3 \mid s$ for every $s \in S$.

Solution. In the language above, we have $b_1 = 3$ and $c(a, b) = a + b$. Proceeding by structural induction, we first need to check the basis cases, which is $3 \mid b_1$. Since $b_1 = 3$ and $3 \mid 3$, the basis case checks out. Now suppose that $a, b \in S$ and that $3 \mid a$ and $3 \mid b$. We want to show that $3 \mid c(a, b)$; that is, $3 \mid (a + b)$. Since $3 \mid a$ and $3 \mid b$, we know there exist $m, n \in \mathbb{Z}$ such that $a = 3m$ and $b = 3n$, so that

$$a + b = 3m + 3n = 3(m + n)$$

showing that $3 \mid (a + b)$ as required. ■

Exercise 17 Consider the following subset S of $\mathbb{Z}^* \times \mathbb{Z}^*$ defined as follows:

- $(0, 0) \in S$
- If $(a, b) \in S$ then $(a, b+1) \in S$, $(a+1, b+1) \in S$, and $(a+2, b+1) \in S$.

Show that $a \leq 2b$ for all $(a, b) \in S$.

Exercise 18 Consider the set of formal strings (characters written without any implied meaning). Define the set M as follows:

- The empty string $\varepsilon \in M$.
- If $a, b \in M$ then $\heartsuit a \clubsuit b \in M$.

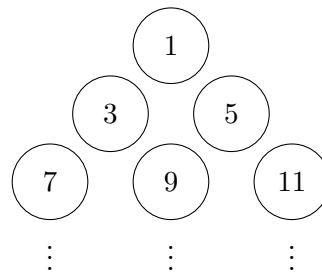
So for example, if $a = \heartsuit \clubsuit$ and $b = \heartsuit \heartsuit \clubsuit \clubsuit$ then

$$\heartsuit a \clubsuit b = \heartsuit \underbrace{\heartsuit \clubsuit}_a \underbrace{\heartsuit \heartsuit \clubsuit \clubsuit}_b$$

Show that for any $m \in M$, the number of $\heartsuit$'s and $\clubsuit$'s are the same.

Induction

1. Show that for any $n \in \mathbb{Z}^+$ we have $\sum_{k=1}^n k = \frac{k(k+1)}{2}$.
2. Show that $\sqrt[n]{2} \leq 1 + \frac{1}{n}$ for all $n \in \mathbb{Z}^+$.
3. Find an expression for $\sum_{k=1}^n (-1)^k (2k-1)^2$ and prove that your expression is correct.
4. Suppose that the odd numbers are arranged in a pyramid, as follows:



Find a formula for the sum of the numbers on the n th row. Prove that your formula is correct.

5. Show that every positive integer $n \in \mathbb{Z}^+$ can be written as a sum of distinct powers of 2; namely, that every integer has a binary representation.
6. Let S be the set of all finite binary strings; that is, the set of strings made up of the numbers 0 and 1, each occurring only finitely many times. Binary strings have the following properties:
 - Given two binary strings $s, t \in S$, we define their *concatenation* st as the binary string formed by appending the digits of t to the end of s . So if $s = 0111$ and $t = 101$ then $st = 0111101$.
 - If $s \in S$, the *length* of the binary string is the number $|s|$ of characters it contains. For example, 0111101 is a binary string of length 7.
 - Given a binary string $s \in S$, we define its *reversal* s^R to be the binary string consisting of its digits in reverse. For example, if $s = 0111101$ above, then $s^R = 1011110$.
 - The *empty string* λ is the string consisting of no characters.

Prove that for any $s, t \in S$, $(st)^R = t^R s^R$.

7. We have learned that logical statements may be built from simple building blocks, such as T for true, F for false, proposition P , and the operations $\neg, \wedge, \vee, \implies, \iff$. We recursively create any valid logical statement as follows:
 - Base Case: Suppose T, F, and any proposition R are valid.
 - Recursive Step: If P and Q are valid formulae, then $(\neg P)$, $(P \wedge Q)$, $(P \vee Q)$, $(P \implies Q)$, and $(P \iff Q)$ are all valid formulae.

Show that any valid logical statement contains an equal number of left- and right-parentheses.